

VFSTR - VADLAMUDI Vadlamudi	Regd. No.													
	Year	Semester			Branch				Sections					
	I	II			CSE,ECE,VLSI				14-25					
	Course: EP													
Program Name: B. Tech.	Date: 10.03.26			Time: 02.00 to 03.30 PM (Each Course 30 Minutes)					Pre-T1					

Time allotted for each course assessment: 30 Minutes.

Max. Marks: 10 Marks

- ❖ Answer all two question from each course
- ❖ Answers should be written for course wise in different sheets

EP (25PY101)

Max. Marks 10

1. Obtain an expression for the Hall voltage in terms of magnetic field density and current. 5M
2. Obtain an expression for the electrical conductivity of conductors based on the classical free electron theory. 5M

what is w.k.t.?

what is $\phi(ev)$ in

$$V_H = \frac{IB}{\phi_{ev}}?$$

(iv) Under steady state condition, (1M)

$$\vec{F}_e + \vec{F}_B = 0$$

$$eE_H - ev_h B_z = 0$$

$$E_H = v_h B_z$$

(v) Hall voltage $\leftarrow$ Hall field

(1M)

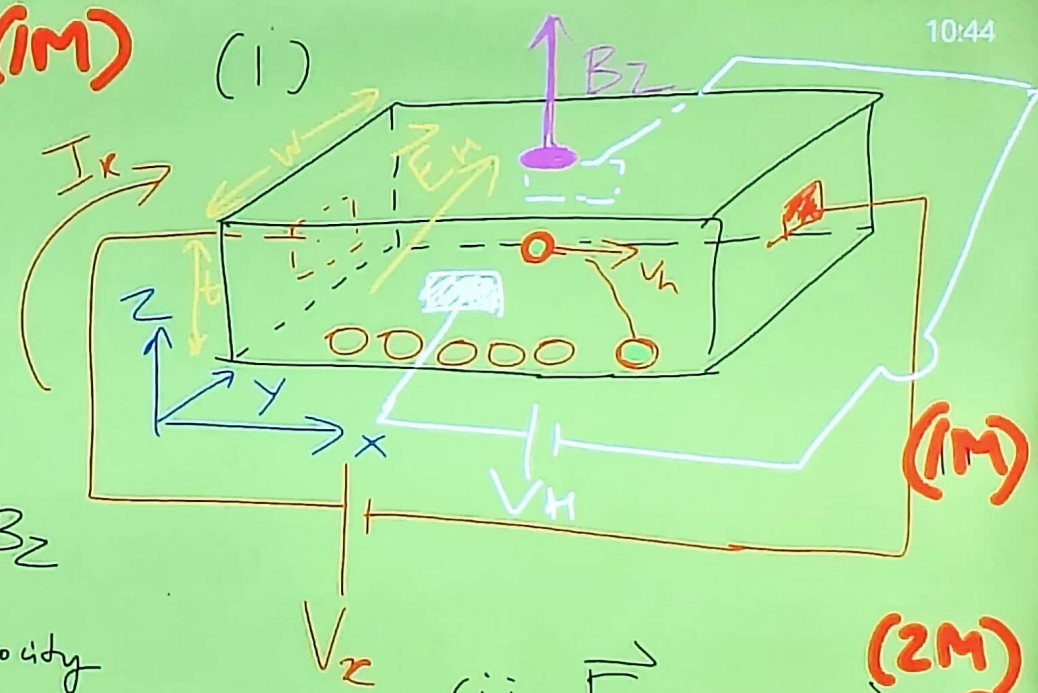
$$V_H = E_H \cdot w = v_h w B_z$$

Current $\leftarrow$

(vi) Current density $\leftarrow$ drift velocity

$$j = pe v_h \Rightarrow v_h = \frac{j}{pe} = \frac{I_x}{pewt}$$

$$(vii) \therefore V_H = \frac{I_x w B_z}{pewt} = \frac{I_x B_z}{pet} = V_H$$



(1M)

$$(ii) \vec{F}_B = +e \vec{v}_h \times \vec{B}_z$$

$$(iii) \vec{F}_E = +e \vec{E}_H = e E_H \hat{j}$$

Hence derived

(2M)

(i) Microscopic version of Ohm's law

(1M)

$$\vec{j} = \sigma \vec{E}$$

$$\Rightarrow \sigma = \frac{j_x}{E_x}$$

(ii) a) Current density $\leftarrow$ Current

(1)

$$j_x = \frac{I_x}{A} = \frac{I_x}{wt}$$

(ii) b) Current $\leftarrow$ Drift velocity

(1)

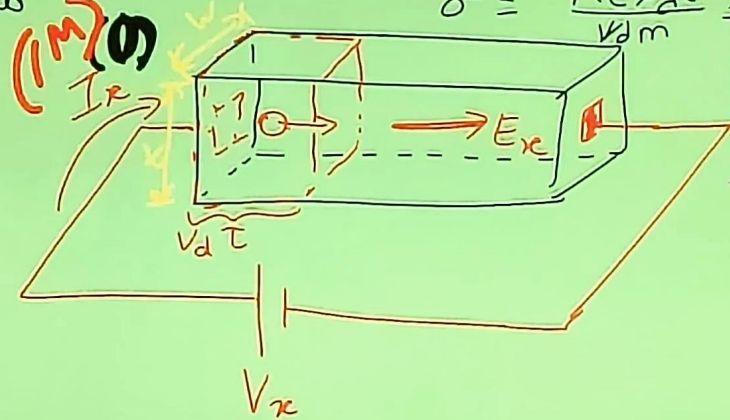
$$I_x = \frac{n \cdot wt \cdot v_d \tau \cdot e}{\tau}$$

a) & b) $\Rightarrow j_x = \frac{newtv_d}{wt\tau} = nev_d$

(iv) Plugging (ii) & (iii) in (i),

$$\sigma = \frac{nev_d \tau}{v_d m} = \frac{ne\tau}{m}$$

10:44



(iii) Electric field $\leftarrow$ Drift velocity

(1)

$$\vec{F}_E = e\vec{E}_x = \vec{F}_E$$

$$\vec{F}_E = m\vec{a}$$

$$\Rightarrow a = \frac{eE_x}{m}$$

$$\Rightarrow v_d = a\tau = \frac{eE_x\tau}{m}$$

$$E_x = \frac{V_d m}{e\tau}$$